



Linear Regression

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Summary from Last Time

Comparison of Two Models

How do we choose from two different models?

Model Fitness

How does the model perform predicting?

Evaluating Significance of Predictors

Does the outcome depend on the predictors?

How well do we know $\hat{f}$

The confidence intervals of our $\hat{f}$



Lecture Outline

- Linear models
- Estimate of the regression coefficients
 - Brute Force
 - Exact method
- Confidence intervals for the predictors estimates
- Bootstrap
- Evaluating significance of predictors
- How well we know the model $\hat{f}$



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Linear Models

Note that in building our kNN model for prediction, we did not compute a closed form for $\hat{f}$.

What if we ask the question:

“how much more sales do we expect if we double the TV advertising budget?”

Alternatively, we can build a model by first assuming a simple form of f :

$$Y = f(X) + \epsilon = \beta_1 X + \beta_0 + \epsilon.$$



Linear Regression

... then it follows that our estimate is:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_1 X + \hat{\beta}_0$$

where $\hat{\beta}_1$ and $\hat{\beta}_0$ are **estimates** of β_1 and β_0 respectively, that we compute using observations.



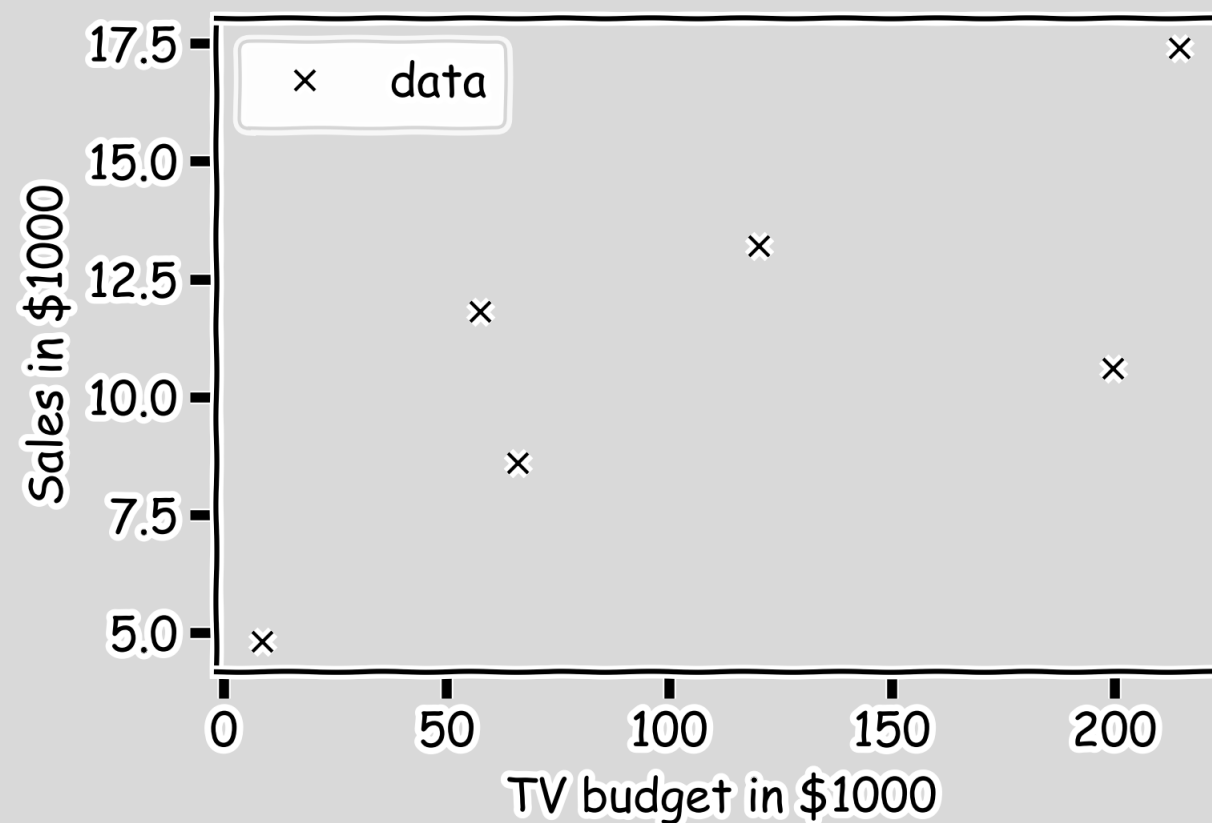
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Estimate of the regression coefficients

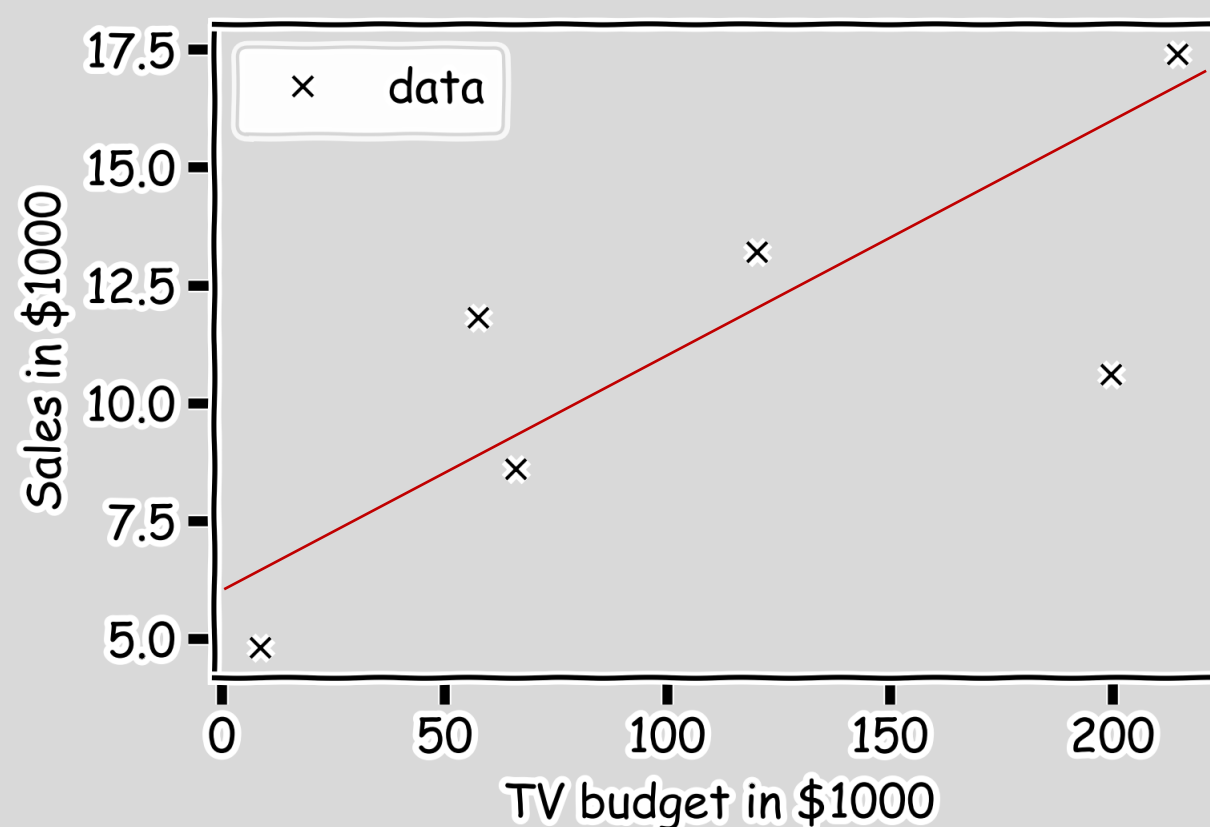
For a given data set





Estimate of the regression coefficients

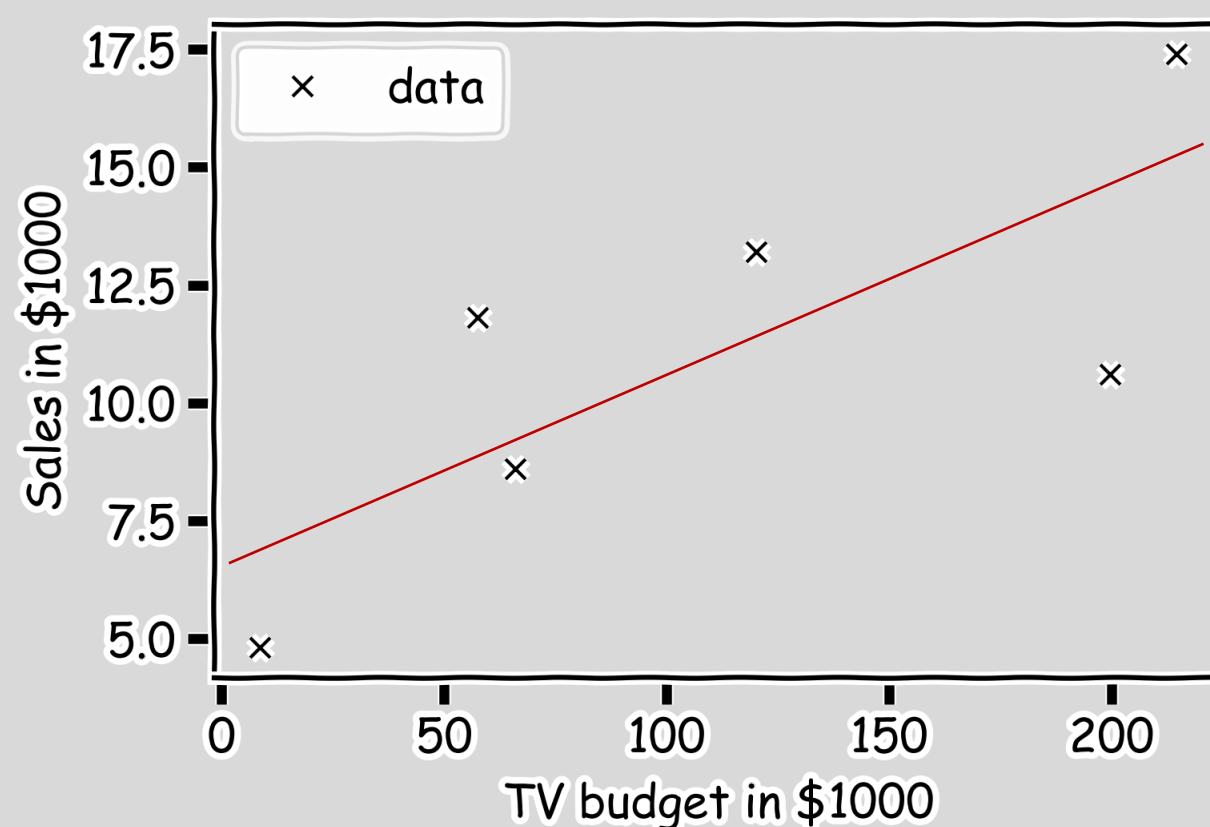
Is this a good line?





Estimate of the regression coefficients

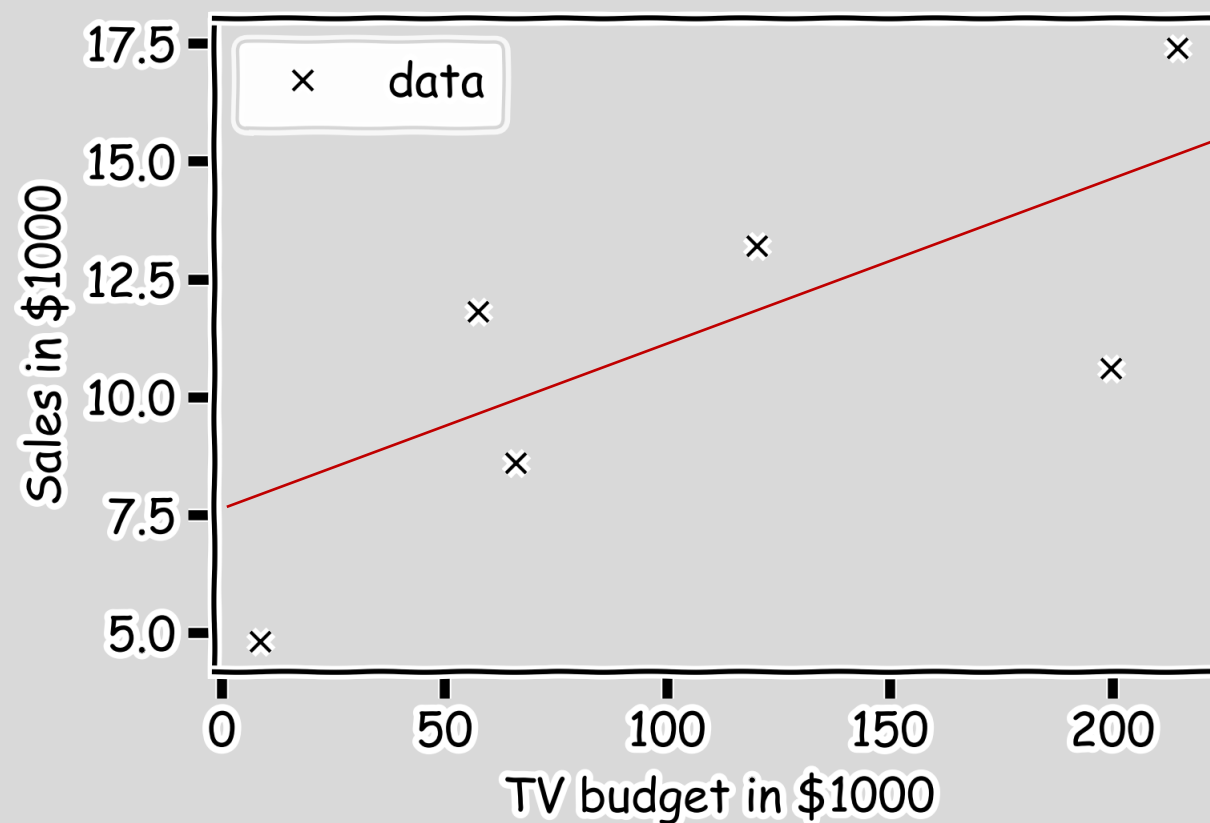
Or maybe this one?





Estimate of the regression coefficients

Or this one?

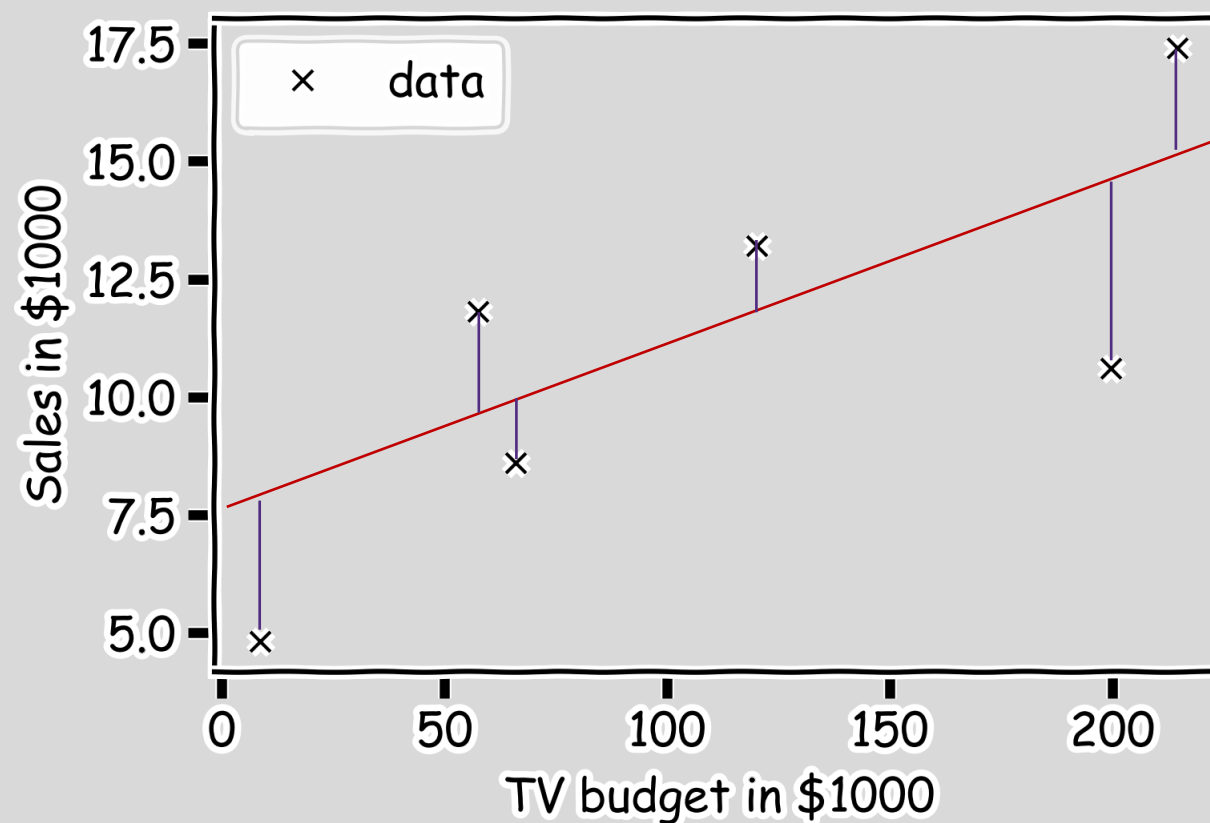




Estimate of the regression coefficients

Question: Which line is the best?

First calculate
the residuals



Estimate of the regression coefficients

Again, we use MSE as our loss function,

$$L(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n [y_i - (\beta_1 X + \beta_0)]^2.$$

We choose $\hat{\beta}_1$ and $\hat{\beta}_0$ in order to minimize the predictive errors made by our model, i.e. minimize our loss function.

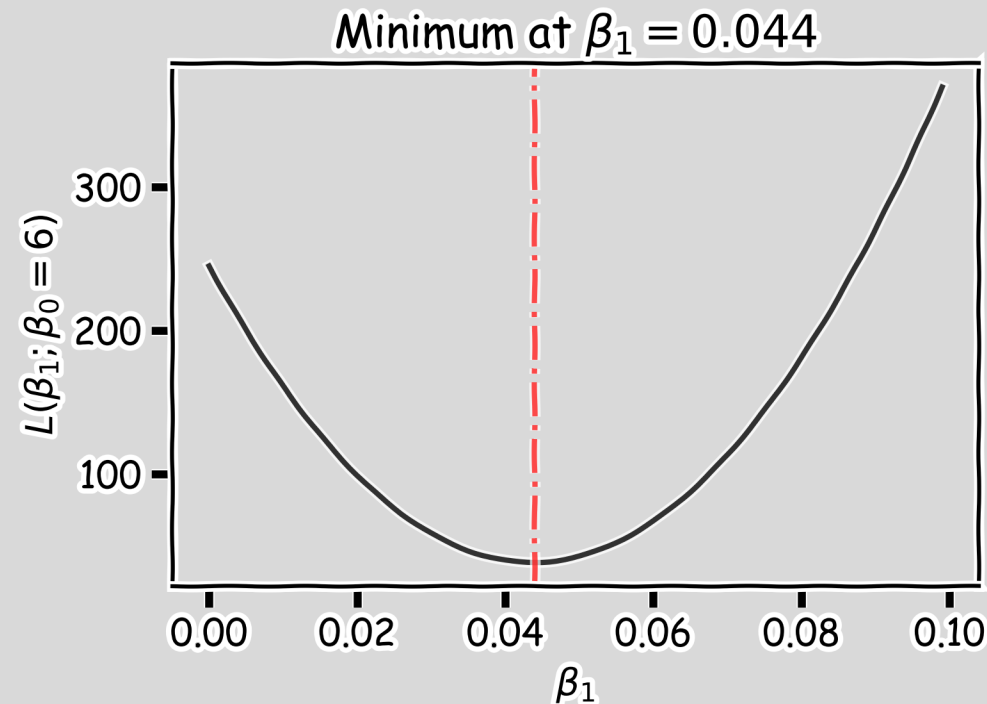
Then the optimal values for $\hat{\beta}_0$ and $\hat{\beta}_1$ should be:

$$\hat{\beta}_0, \hat{\beta}_1 = \underset{\beta_0, \beta_1}{\operatorname{argmin}} L(\beta_0, \beta_1).$$

Estimate of the regression coefficients: brute force

A way to estimate $\operatorname{argmin}_{\beta_0, \beta_1} L$ is to calculate the loss function for every possible β_0 and β_1 . Then select the β_0 and β_1 where the loss function is minimum.

E.g., the loss function for different β_1 when β_0 is fixed to be 6:



Estimate of the regression coefficients: Exact Method

Take the partial derivatives of L with respect to β_0 and β_1 , set to zero, and find the solution to that equation. This procedure will give us explicit formulae for $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{y}$ and $\bar{x}$ are sample means.

The line:

$$\hat{Y} = \hat{\beta}_1 X + \hat{\beta}_0$$

is called the **regression line**.

Estimate of the regression coefficients: Exact Method

$$\frac{dL(\beta_0, \beta_1)}{d\beta_0} = 0$$

$$\Rightarrow \frac{2}{n} \sum_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\Rightarrow \frac{1}{n} \sum_i y_i - \beta_0 - \beta_1 \frac{1}{n} \sum_i x_i = 0$$

$$\Rightarrow \beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\frac{dL(\beta_0, \beta_1)}{d\beta_1} = 0$$

$$\Rightarrow \frac{2}{n} \sum_i (y_i - \beta_0 - \beta_1 x_i)(-x_i) = 0$$

$$\Rightarrow - \sum_i x_i y_i + \beta_0 \sum_i x_i + \beta_1 \sum_i x_i^2 = 0$$

$$\Rightarrow - \sum_i x_i y_i + (\bar{y} - \beta_1 \bar{x}) \sum_i x_i + \beta_1 \sum_i x_i^2 = 0$$

$$\Rightarrow \beta_1 \left(\sum_i x_i^2 - n \bar{x}^2 \right) = \sum_i x_i y_i - n \bar{x} \bar{y}$$

$$\Rightarrow \beta_1 = \frac{\sum_i x_i y_i - n \bar{x} \bar{y}}{\sum_i x_i^2 - n \bar{x}^2}$$

$$\Rightarrow \beta_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$



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Interpretation of Predictors

Question: What do you think a predictor coefficient means?

$$Sales = 7.5 + 0.04 TV$$

What does 7.5 mean and what does 0.04 mean?

If we increase the TV by \$1000, what would you expect the increase in sales to be?

What if?

$$Sales = 7.5 + 1.01 TV$$

The interpretation of the predictors depends on the values, but decisions depend on how much we trust these values.

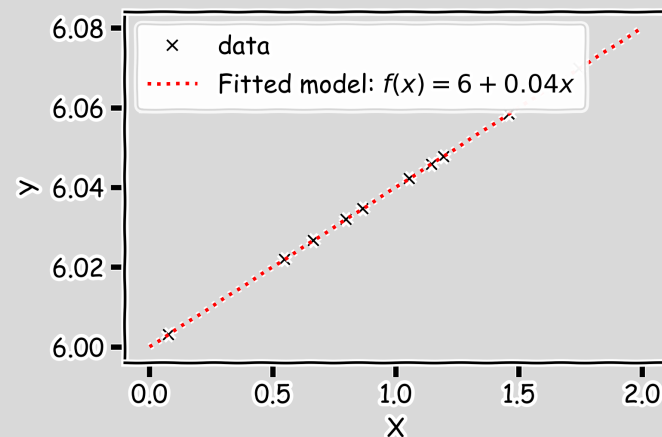
Confidence intervals for the predictors estimates

We interpret the ε term in our observation

$$y = f(x) + \varepsilon$$

to be noise introduced by random variations in natural systems or imprecisions of our scientific instruments.

If we knew the exact form of $f(x)$, for example, $f(x) = \beta_0 + \beta_1 x$, and there was no ε , then estimating the $\hat{\beta}'s$ would have been exact (so is 1.01 worth it?).



Confidence intervals for the predictors estimates

However, three things happen, which result in mistrust of the values of $\hat{\beta}'s$:

- ε is always there
- we do not know the exact form of $f(x)$
- limited sample size

We will first address ε

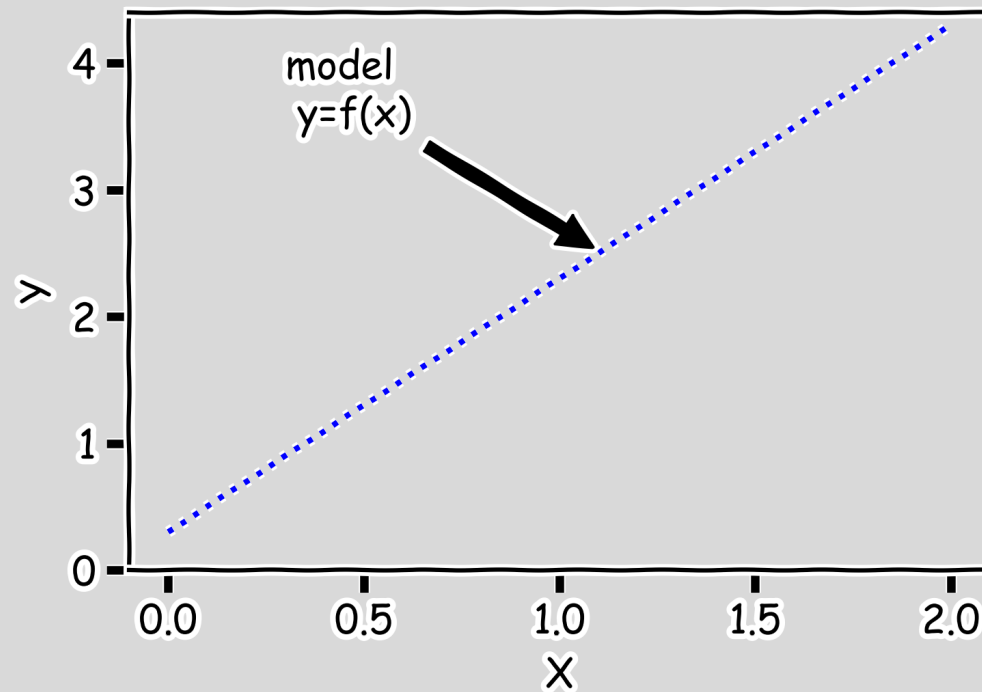
We call ε the measurement error or *irreducible error*. Since even predictions made with the actual function f will not match observed values of y .

Because of ε , every time we measure the response Y for a fix value of X , we will obtain a different observation, and hence a different estimate of $\hat{\beta}'s$.



Confidence intervals for the predictors estimates

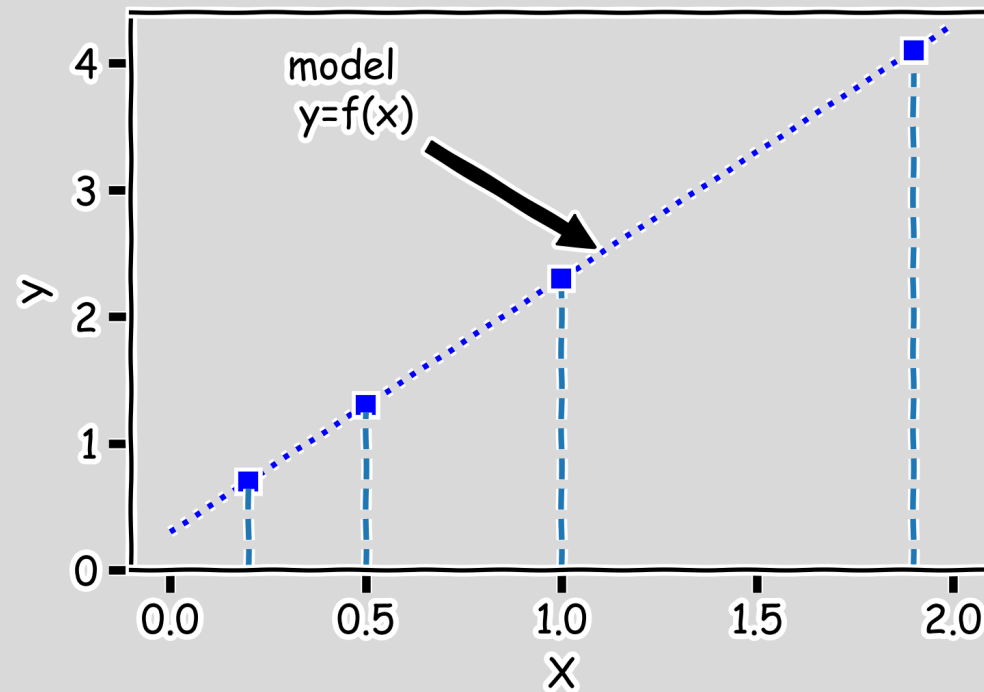
Start with a model





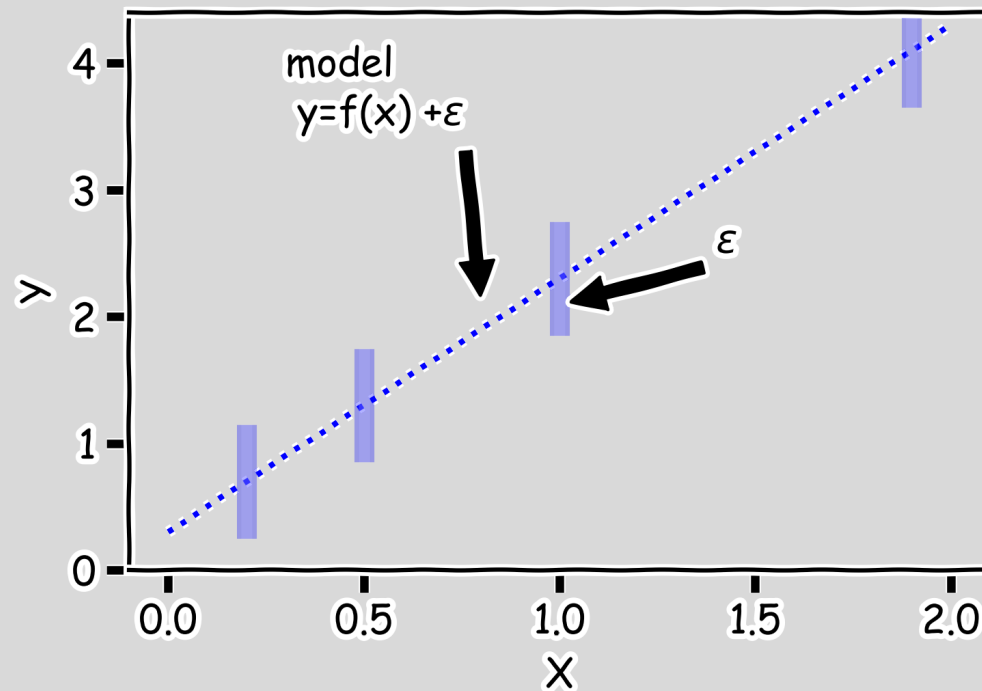
Confidence intervals for the predictors estimates

For some values of X , $Y = f(X)$



Confidence intervals for the predictors estimates

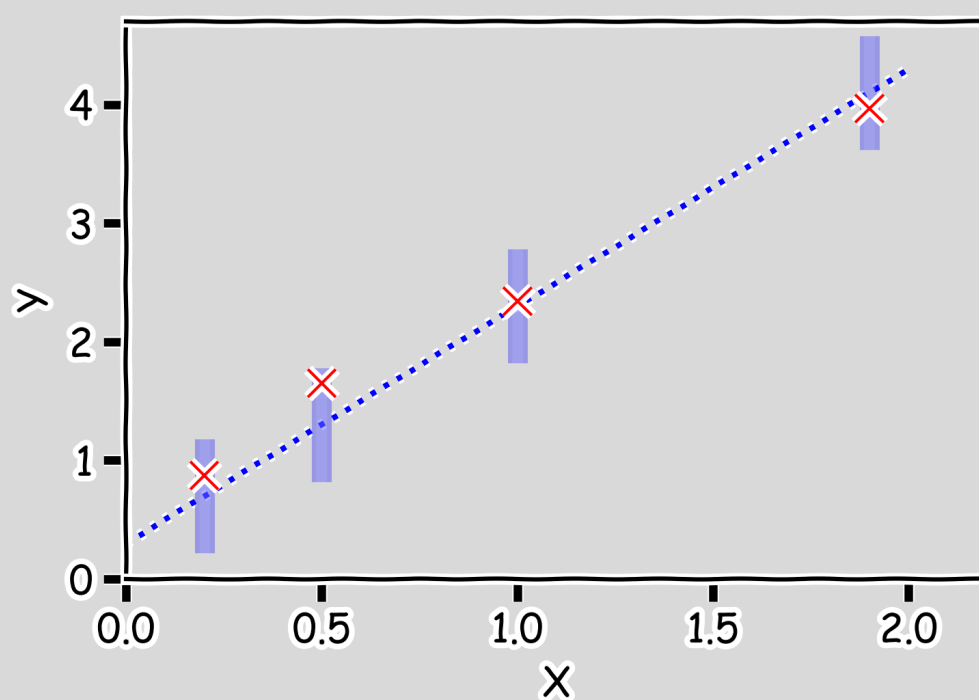
But due to error, every time we measure the response Y for a fixed value of X we will obtain a different observation.





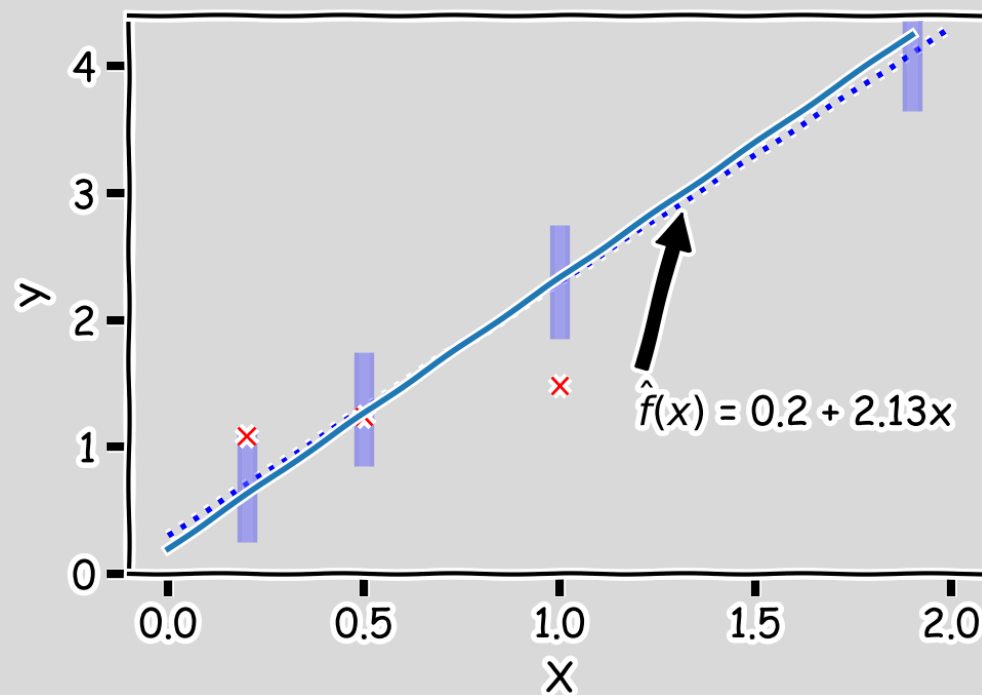
Confidence intervals for the predictors estimates

One set of observations, “one realization” we obtain one set of Y s (red crosses).



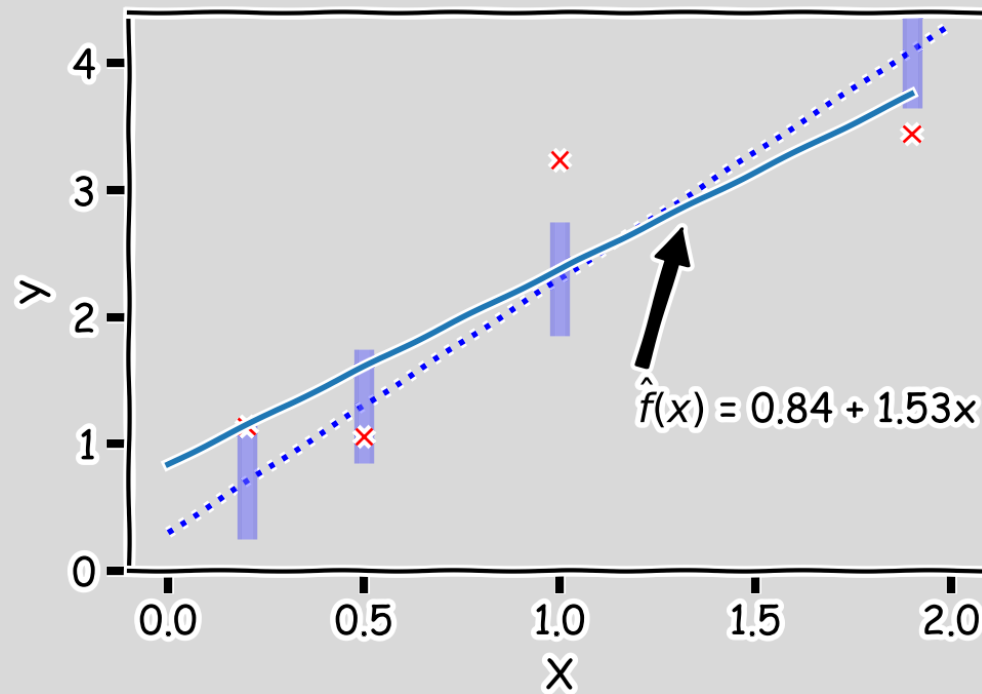
Confidence intervals for the predictors estimates

For each one of those “realizations”, we could fit a model and estimate $\hat{\beta}_0$ and $\hat{\beta}_1$.



Confidence intervals for the predictors estimates

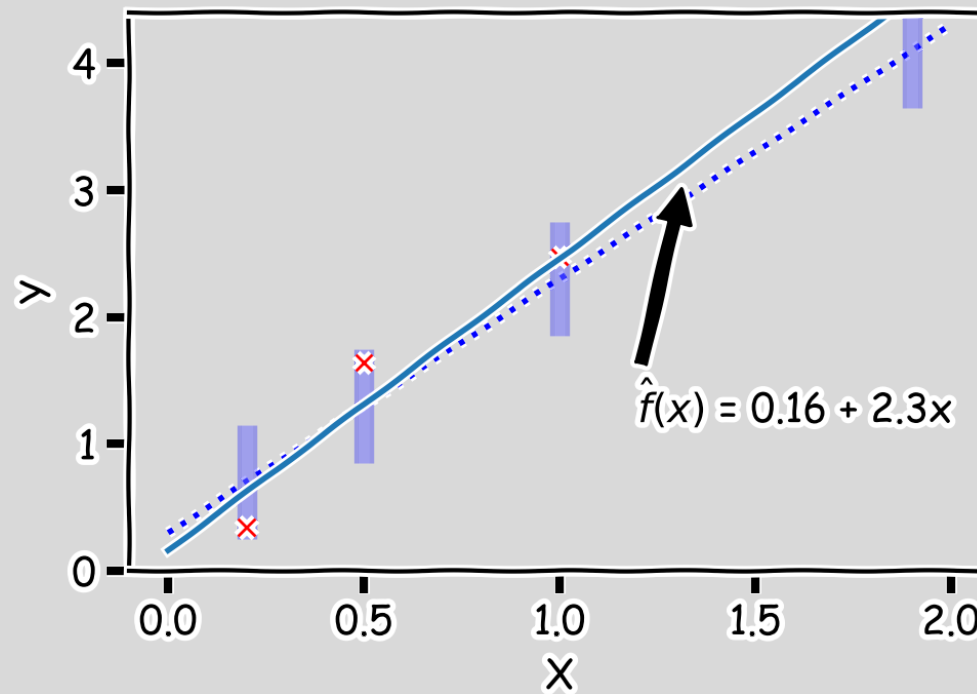
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Confidence intervals for the predictors estimates

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Confidence intervals for the predictors estimates

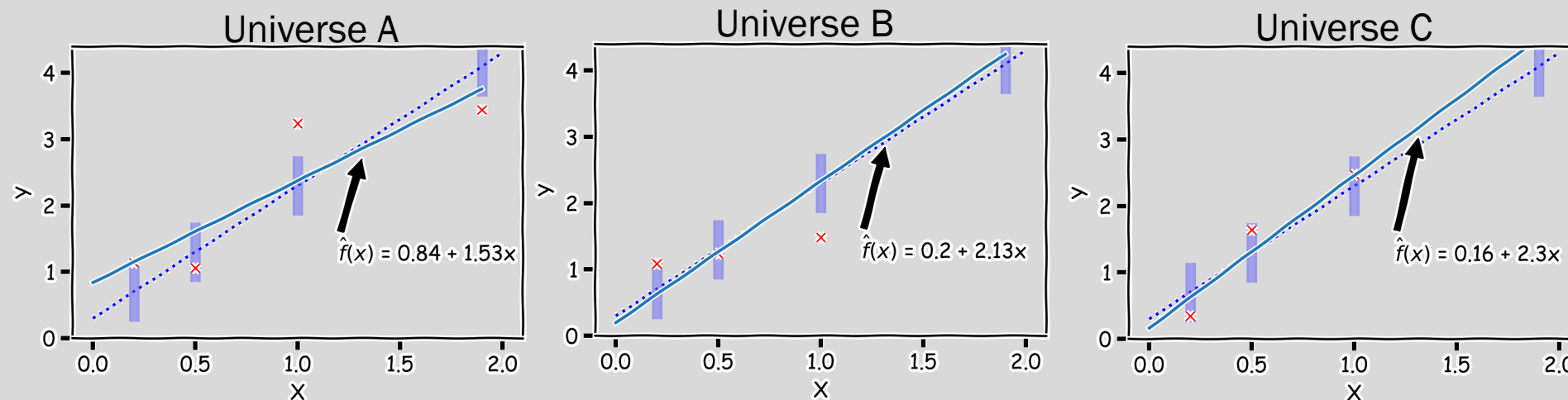
So if we just have one set of measurements of $\{X, Y\}$, our estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$ are just for this particular realization.

Confidence intervals for the predictors estimates

So if we just have one set of measurements of $\{X, Y\}$, our estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$ are just for this particular realization.

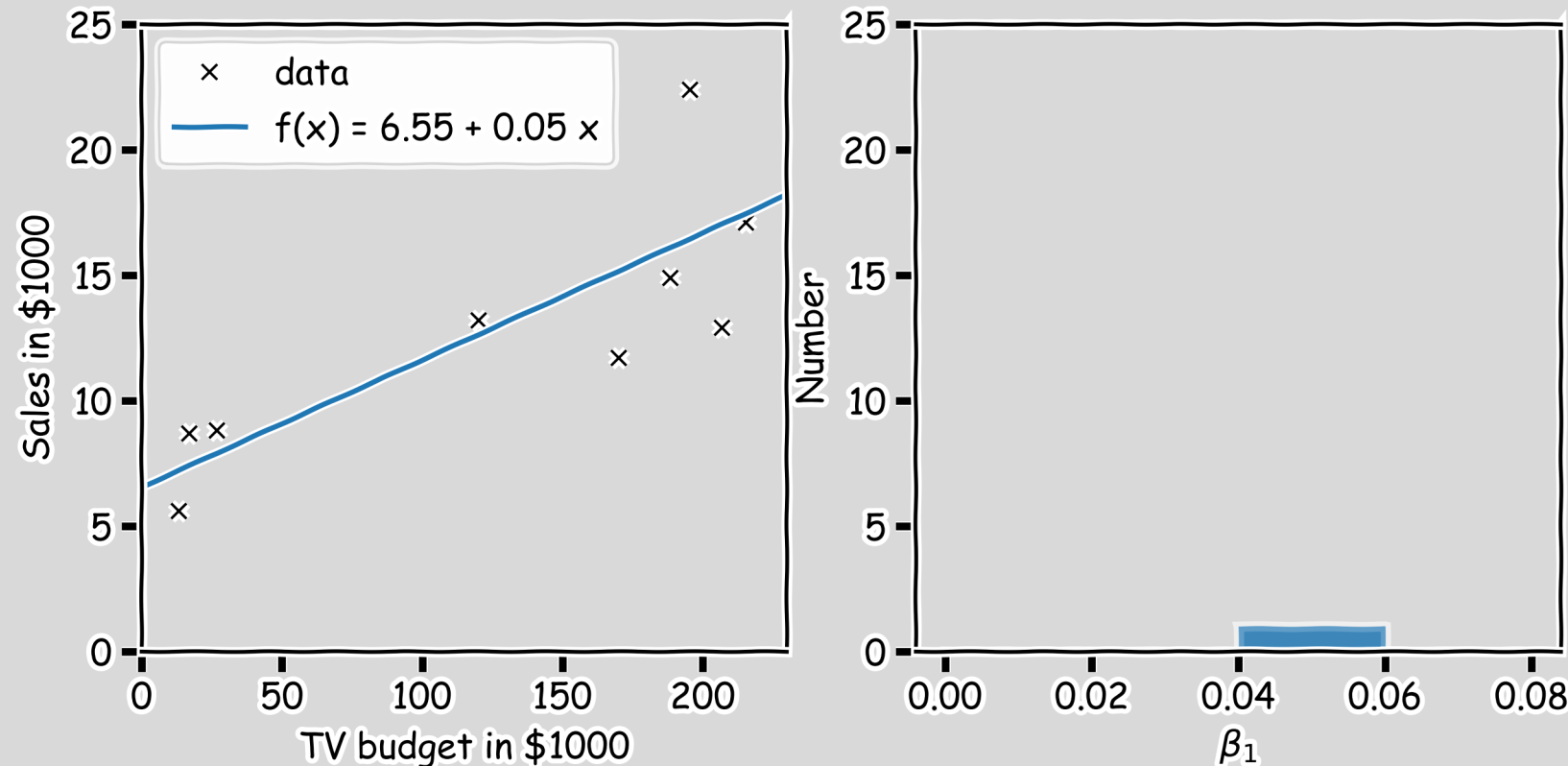
Question: If this is just one realization of the reality how do we know the truth? How do we deal with this conundrum?

Imagine (magic realism) we have parallel universes and we repeat this experiment on each of the other universes.



Confidence intervals for the predictors estimates

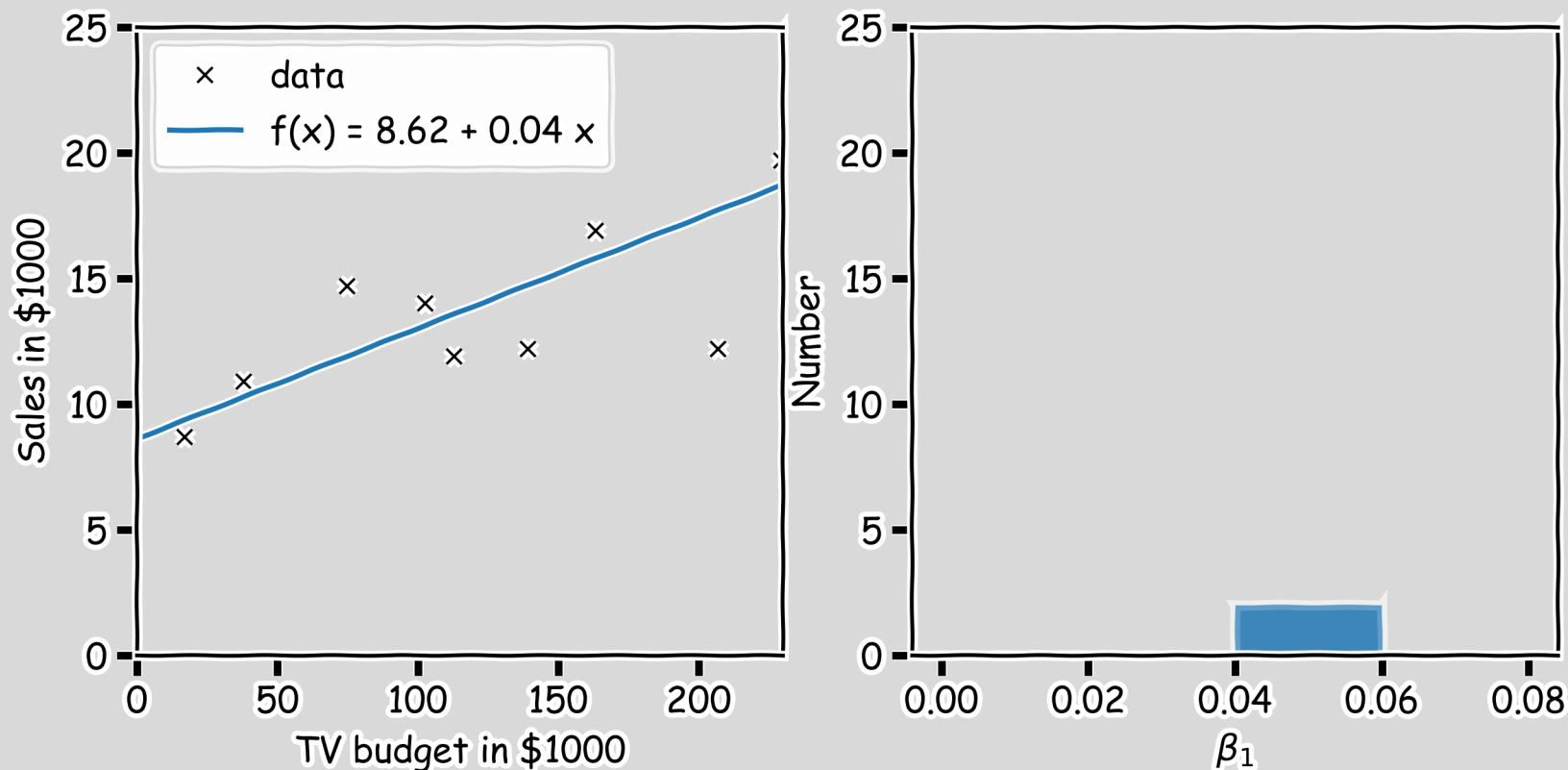
In our magical realisms, we can now sample multiple times





Confidence intervals for the predictors estimates

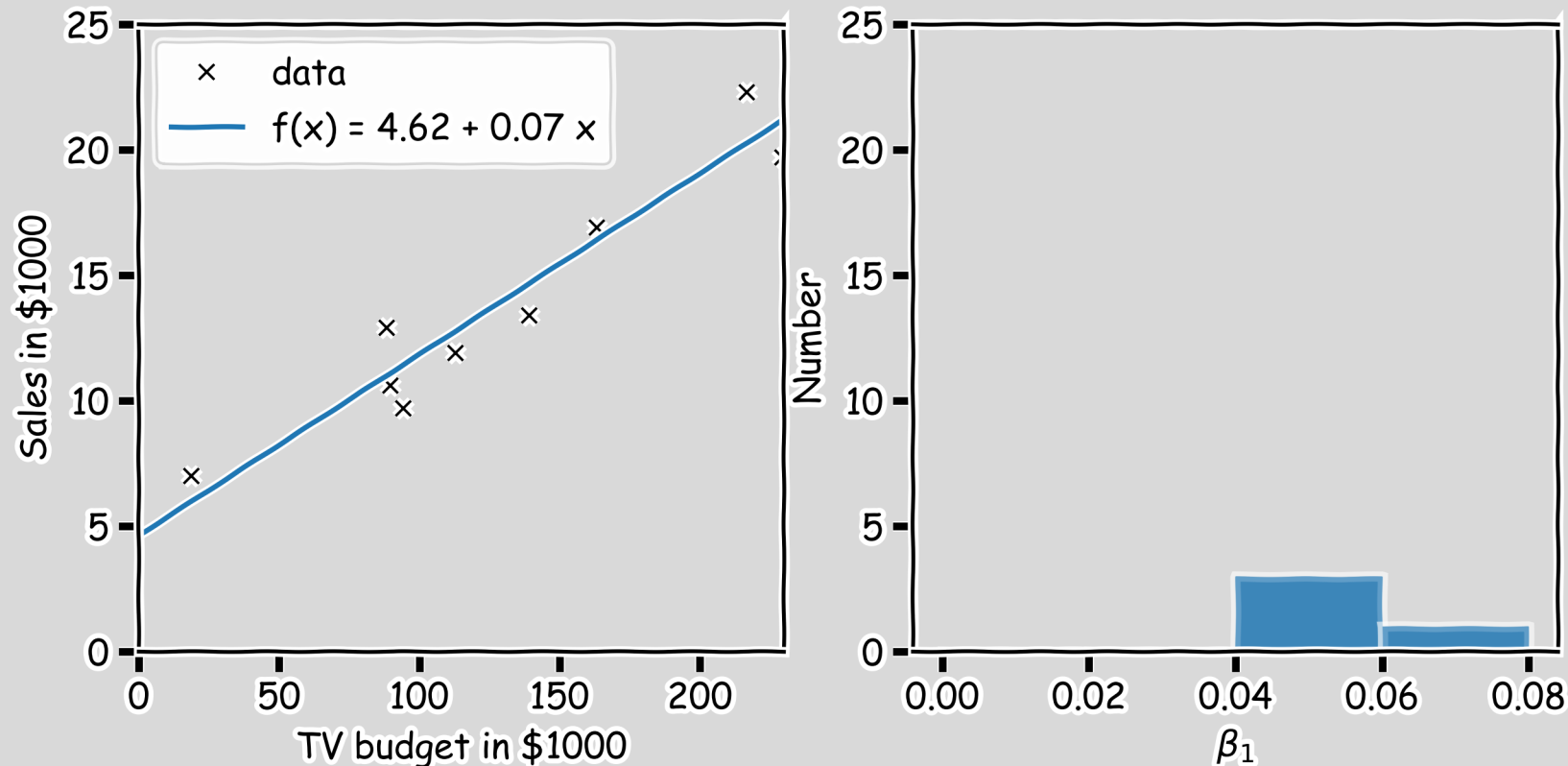
Another sample





Confidence intervals for the predictors estimates

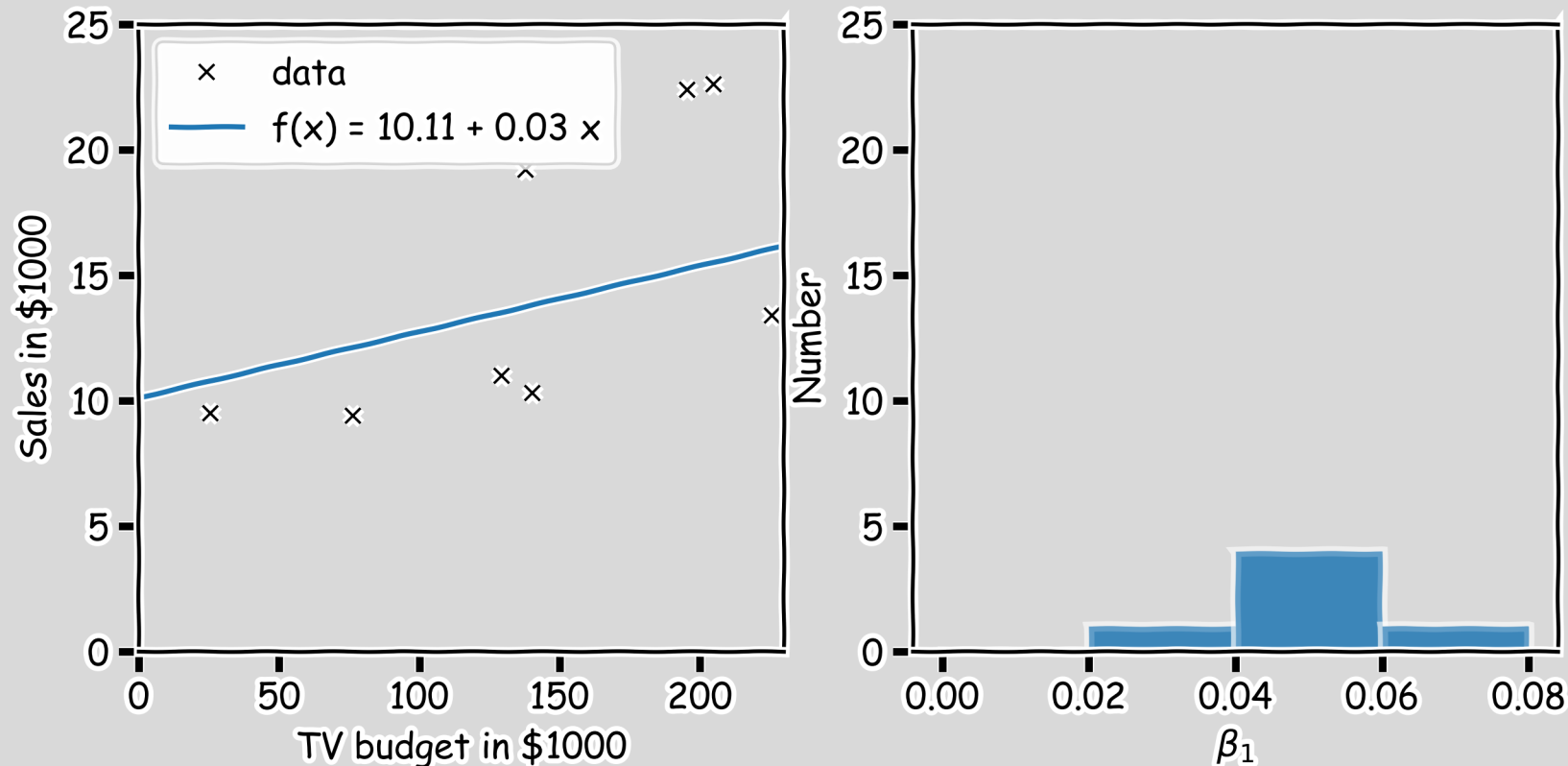
Another sample





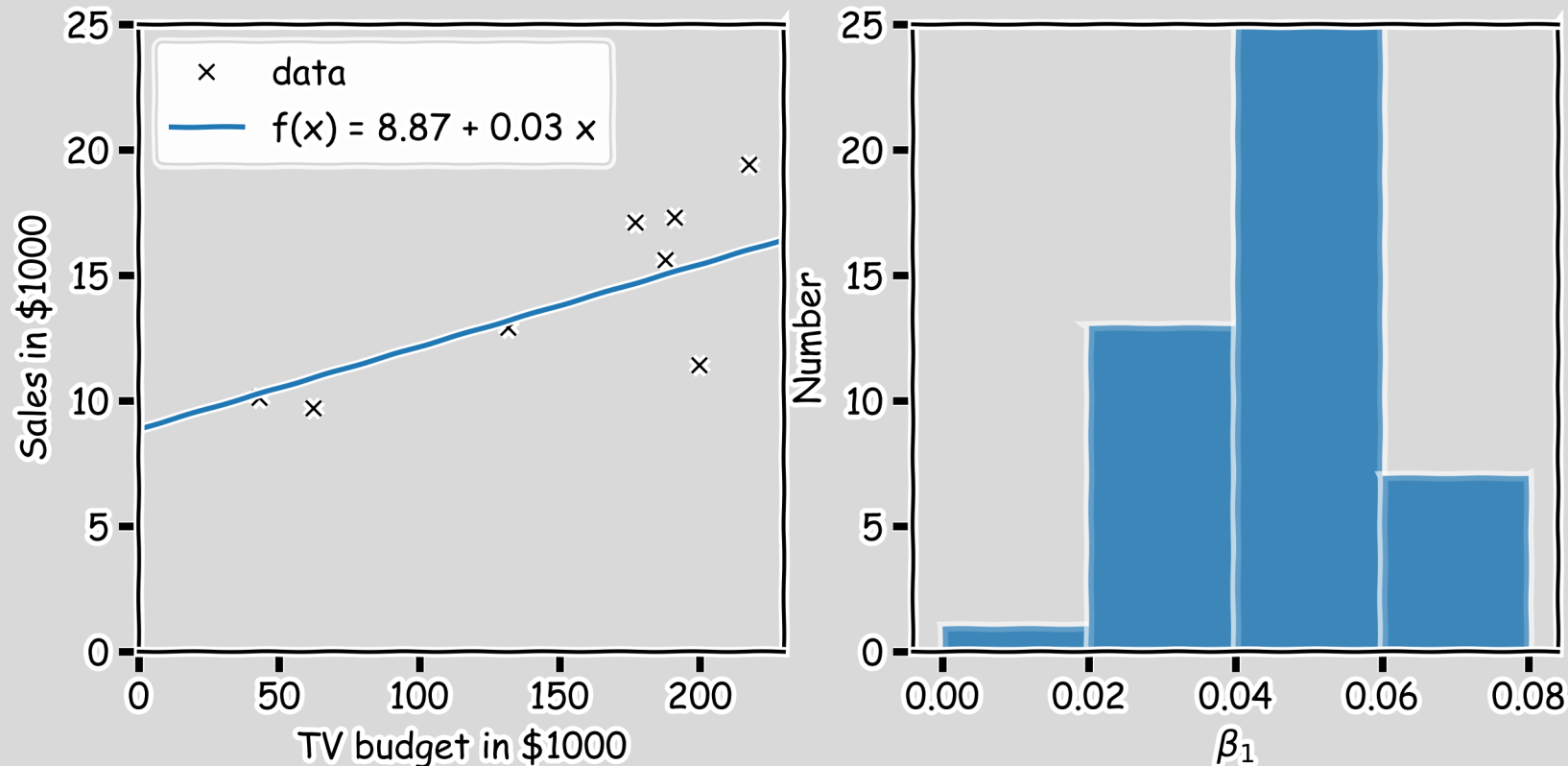
Confidence intervals for the predictors estimates

And another sample



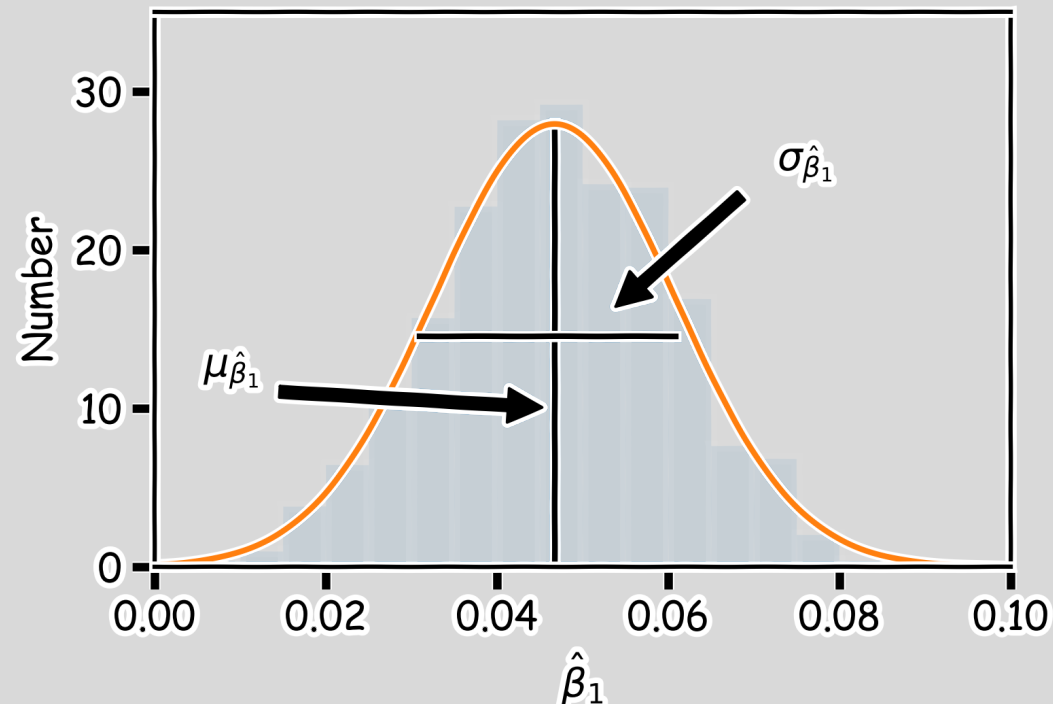
Confidence intervals for the predictors estimates

Repeat this 100 times



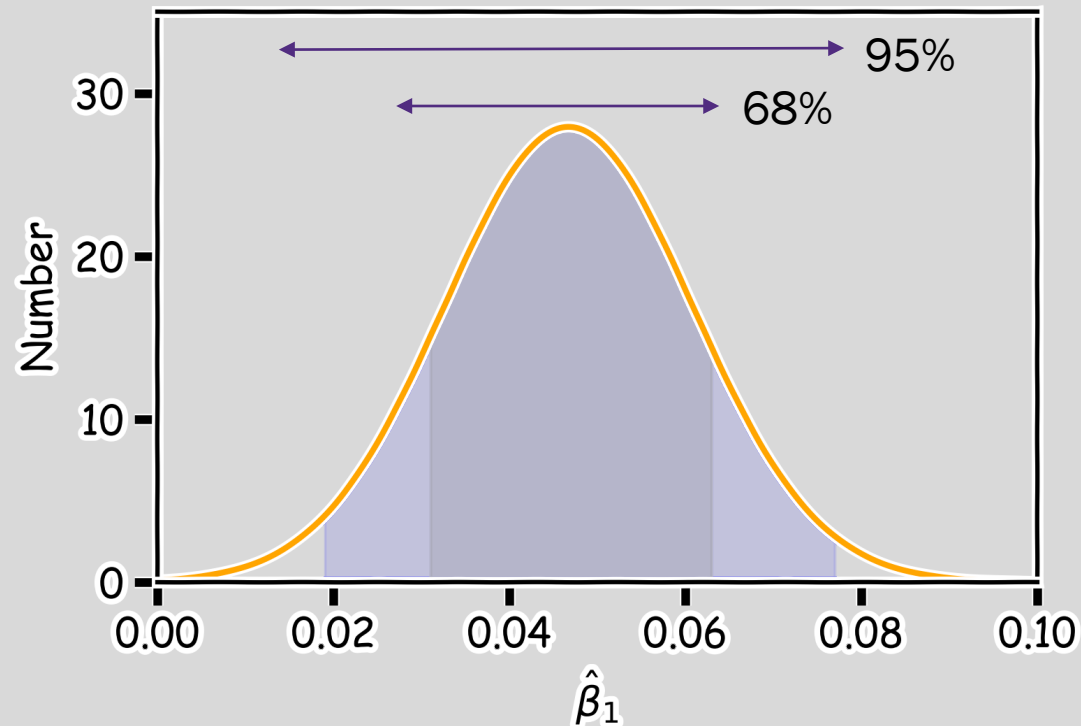
Confidence intervals for the predictors estimates

We can now estimate the mean and standard deviation of all the estimates $\hat{\beta}_1$.
The variance of $\hat{\beta}_0$ and $\hat{\beta}_1$ are also called their *standard errors*, $SE(\hat{\beta}_0)$, $SE(\hat{\beta}_1)$.



Confidence intervals for the predictors estimates

Finally we can calculate the confidence intervals, which are the ranges of values such that the **true** value of β_1 is contained in this interval with n percent probability.



Confidence intervals for the predictors estimates

And also we can answer the question, 'how significant are the predictors?' Here we show the same analysis for all three predictors.

Question: Which ones are important?



Before we answer this question, we need to answer another question.